Slip 08

Q1) A Define Steganography and distinguish it clearly fromCryptography

Q1) A) Define Steganography and distinguish it clearly from Cryptography. (8 Marks)

1. Definition of Steganography

Steganography is the technique of hiding secret information inside an ordinary file, such as an image, audio, video, or text file, so that the existence of the secret message remains hidden.

Example: Hiding a secret message inside a photograph and sending the photograph to another person.

2. Definition of Cryptography

Cryptography is the technique of converting readable information (plaintext) into an unreadable form (ciphertext) to protect it from unauthorized access.

Example: Converting the message `HELLO` into an encrypted message such as `KHOOR` using a Caesar cipher with a key of 3.

3. Difference Between Steganography and Cryptography

Steganography	Cryptography
1. It hides the existence of a message.	1. It hides the meaning of a message.
2. Secret data is embedded inside an ordinary file.	2. Data is converted into ciphertext using an algorithm.

3. The carrier file may look normal.	3. The encrypted message may look unreadable or meaningless.
4. It uses images, audio, videos, or text as carriers.	4. It uses encryption algorithms and keys.
5. Its main goal is to conceal communication.	5. Its main goal is to protect information from unauthorized reading.
6. Example: Hiding a message inside an image.	6. Example: Encrypting a message using AES.
7. If the hidden message is discovered, it may be readable unless separately encrypted.	7. The message generally remains unreadable without the required decryption key.
8. Detection focuses on finding hidden information.	8. Attacks may attempt to break the encryption or obtain the key.

4. Simple Diagram

Steganography

Secret message + Image
Image containing the hidden message

Cryptography

Plaintext: HELLO

Encryption using key = 3

Ciphertext: KHOOR

Conclusion

Steganography hides the existence of a secret message, whereas cryptography hides the meaning of a message. Both techniques help protect information, and they can also be used together for stronger security.

Q2) B Write a Python program that accepts a sentence from the user and

encrypts it using Caesar Cipher

```
text = input("Enter a sentence: ")
key = int(input("Enter key: "))
enc = ""

for ch in text:
    if ch.isalpha():
        enc = enc + chr((ord(ch.upper()) - 65 + key) % 26 + 65)
    else:
        enc = enc + ch

print("Encrypted text:", enc)
```

OR

Q2) B Write a Python program to implement a dictionary attack using a list of common passwords to check whether a target password
exists in the list

```
# Dictionary Attack Simulation

def dictionary_attack(target):
    passwords = [
        "admin123", "password", "123456", "qwerty",
        "welcome123", "computer@123", "hello123",
        "student123", "india@123", "python123"
    ]
```

```
    attempts = 0

    for pwd in passwords:
        attempts += 1
        print("Attempt", attempts, ":", pwd)

        if pwd == target:
            print("Password matched!")
            print("Total attempts:", attempts)
            return

    print("Password not found in dictionary.")
    print("Total attempts:", attempts)


target = input("Enter target password: ")
dictionary_attack(target)
```